SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY

Department of Artificial Intelligence and Data Science

Python Bootcamp Capstone Project

- Problem Statement ID: 24
- Problem Statement Title: Energy - Grid Peak-Load Prediction
- PS Domain: Energy
- Team ID: 24
- Team Members:
  - 24171: Dhruo Suresh Gadhiya
  - 24173: Sameep Dilipkumar Makwana

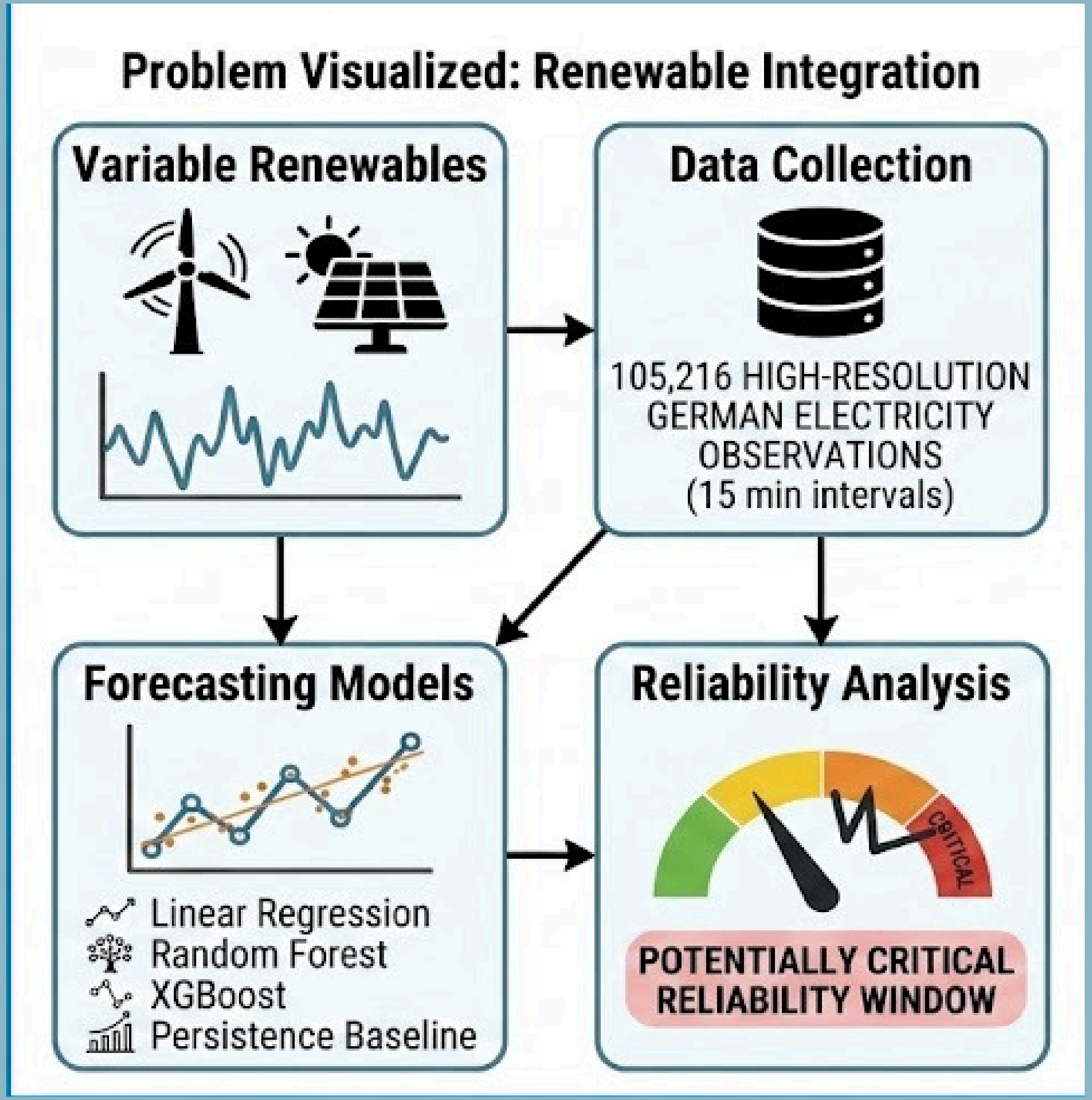
Problem Definition

The increasing variability of renewable energy sources such as solar and wind makes it challenging to maintain a balance between electricity demand and renewable generation.

This project uses 105,216 high-resolution German electricity observations at 15-minute intervals to develop a data-driven framework for short-term load forecasting and renewable reliability analysis.

Linear Regression, Random Forest, XGBoost and a Persistence baseline are evaluated using chronological train–test splitting, while solar, wind and hydro generation are combined to calculate renewable coverage and generation gap.

By integrating demand forecasting with demand–renewable condition analysis, the framework identifies periods of high demand with low renewable coverage as potentially critical reliability windows, providing a data-driven basis for renewable integration and future energy-system planning.



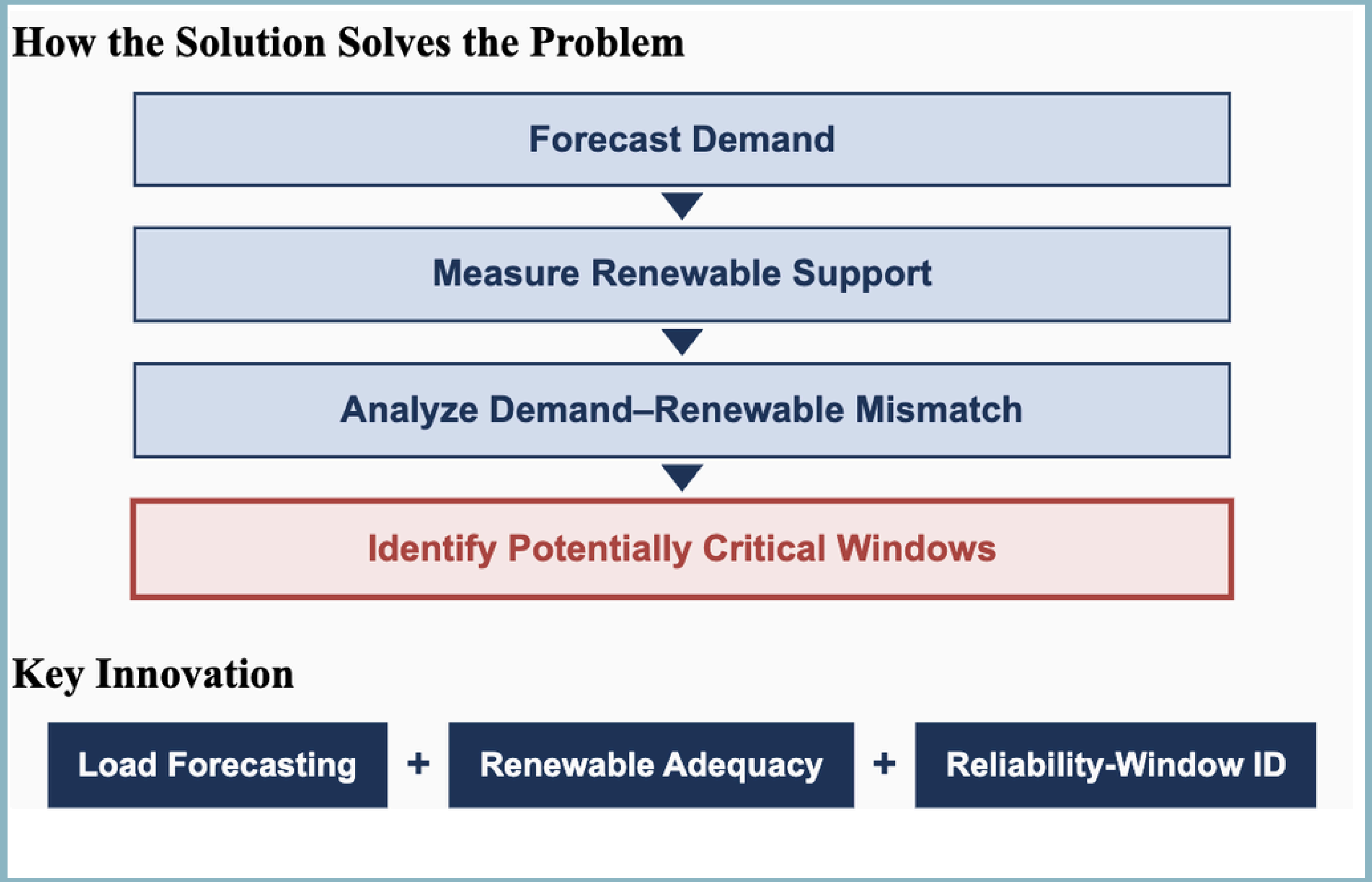
Smart Electricity Demand and Renewable Reliability Analysis

Proposed Solution

A data-driven framework combining short-term electricity-load forecasting with renewable-generation analysis to identify potentially critical periods where high demand coincides with comparatively low renewable support.

Detailed Breakdown

- Validate and preprocess high-resolution German electricity data
- Engineer temporal & historical load features
- Forecast load using multiple ML models
- Compare ML models against a persistence baseline
- Analyze solar, wind and hydro generation
- Calculate renewable coverage and generation gap
- Classify demand–renewable conditions
- Identify potentially critical reliability windows



TECHNICAL APPROACH

Technologies Involved

Python | Pandas | NumPy | Scikit-learn | XGBoost | Matplotlib

Dataset

105,216	22	15-min	3 yrs
Observations	Variables	Resolution	Coverage

Implementation Methodology

Data Validation → Preprocessing → Feature Engineering

Load Forecasting → Renewable Analysis → Reliability Analysis → Model Evaluation

Feature Engineering

Temporal
Hour | Day | Month | Weekday

Historical
Load Lag Features · Rolling Statistics

Experimental Setup

Chronological Train/Test Split
Training: **83,635** observations
Testing: **20,909** observations

Models

Random Forest | XGBoost | Linear Regression | Persistence (Baseline)

FEASIBILITY AND VIABILITY

Feasibility Analysis

Data

- High-resolution historical electricity data
- 105,216 observations · 15-min resolution
- Complete validated dataset

Technical

- Python-based implementation
- Standard machine-learning libraries
- Reproducible processing pipeline
- Standard computing resources

Practical Viability

- Short-term electricity-demand assessment
- Renewable adequacy analysis
- Potentially critical-period identification
- Energy-system planning support

Challenges & Risks

- Strong temporal dependence
- Renewable variability
- Data-quality issues
- Target leakage
- Forecast uncertainty

Mitigation Strategies

- Dataset validation
- Timestamp continuity checks
- Chronological train/test split
- Exclusion of target-derived quantities from forecasting features
- Persistence baseline & multiple evaluation metrics

IMPACT AND BENEFITS

Impact on Target Audience

Grid Planners	Energy Analysts	Renewable Planners	Researchers

Key Benefits

FORECASTING
Short-term electricity-demand estimation

RENEWABLES
Quantification of renewable support relative to demand

RELIABILITY
Identification of potentially critical demand–renewable periods

PLANNING
Support for renewable integration and energy-system analysis

Overall Impact

Load Forecasting + Renewable Availability

↓

Project Outcome

105,216 obs + XGBoost best + 194 windows

Load Forecasting + Renewable Adequacy → Better Demand–Renewable Awareness

RENEWABLE ADEQUACY & DEMAND–RENEWABLE CONDITIONS

Total Renewable Generation

$$R_t = Solar_t + Wind_t + Hydro_t$$

Renewable Coverage Ratio

$$RCR_t = (R_t / L_t) \times 100$$

Renewable Generation Gap

$$RGG_t = L_t - R_t$$

Demand–Renewable Condition Matrix

RENEWABLE AVAILABILITY

	HIGH	LOW
LOW DEMAND	Favorable	Lower Demand Pressure
HIGH DEMAND	Renewable Support	POTENTIALLY CRITICAL

Result: 194 potentially critical reliability windows identified.
Analytical reliability windows derived from demand–renewable conditions (not confirmed grid failures or outages).

EXPERIMENTAL RESULTS

Model Comparison · units: MW for MAE / RMSE

Model	MAE	RMSE	R^2
Persistence	506.90	670.42	0.994445
Linear Regression	289.44	398.98	0.998033
Random Forest	272.41	364.91	0.998354
XGBoost (Best)	**254.36**	**341.28**	**0.998561**

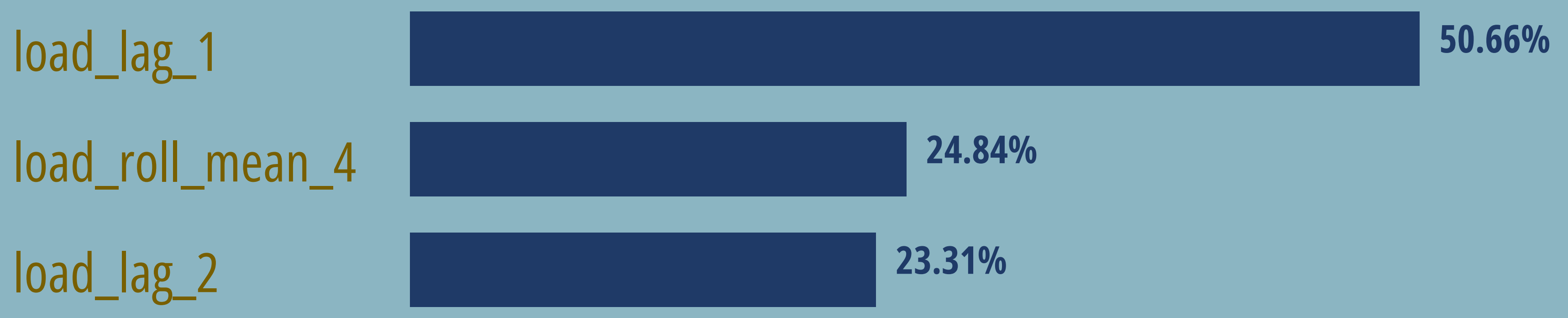
BEST-PERFORMING MODEL
XGBoost

Best MAE
254.36

Best RMSE
341.28

Best R^2
0.9986

XGBoost Feature Importance



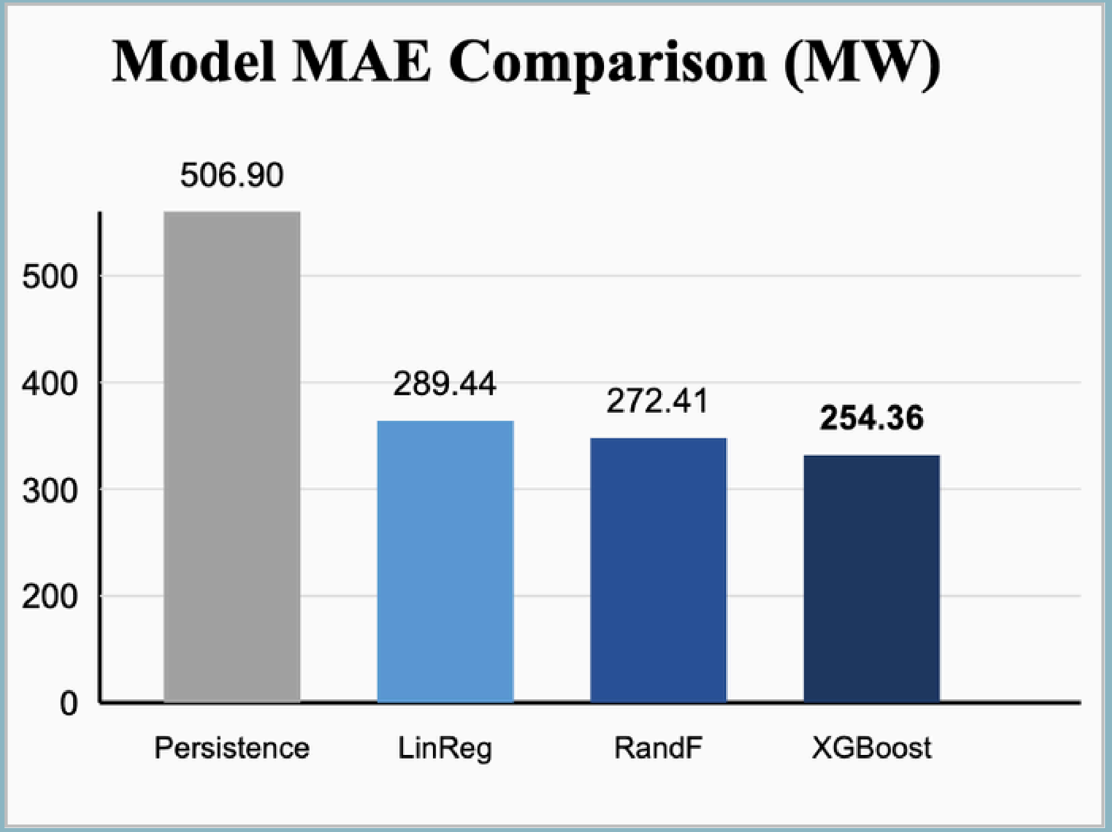
Recent historical load is the dominant source of predictive information.

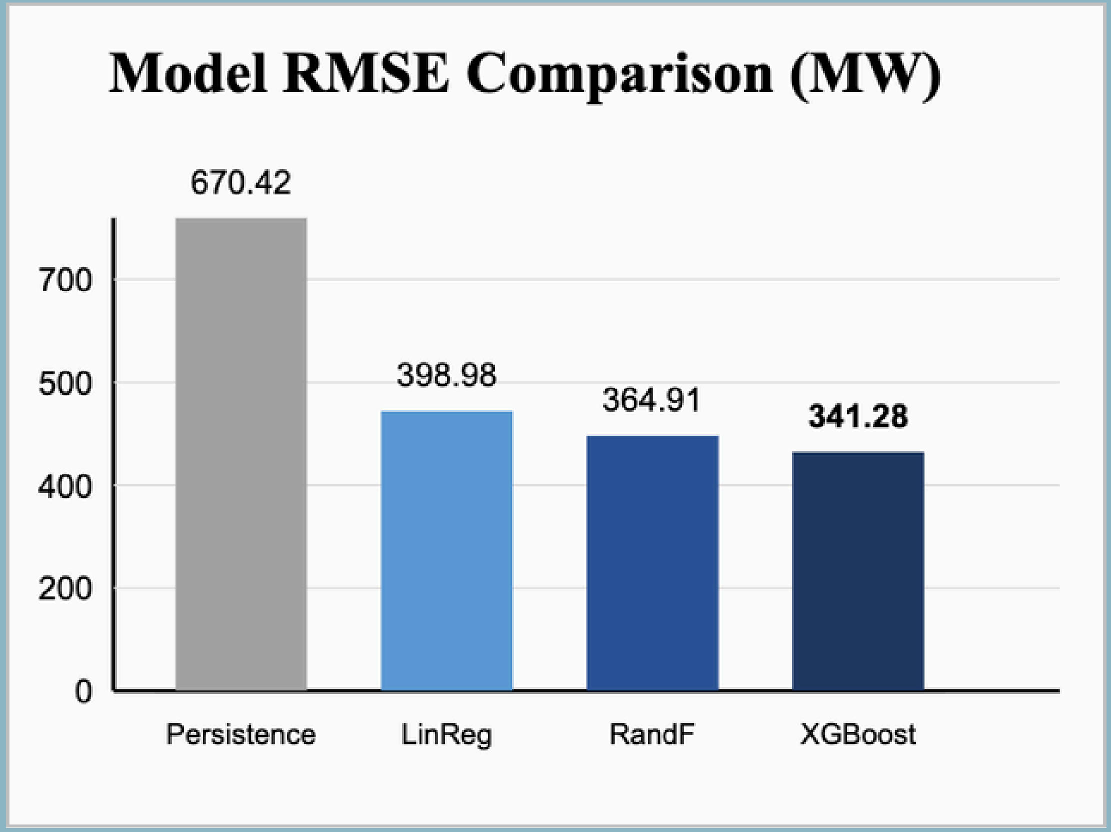
Persistence MAE 506.90 → XGBoost MAE 254.36
≈ 49.8% MAE reduction

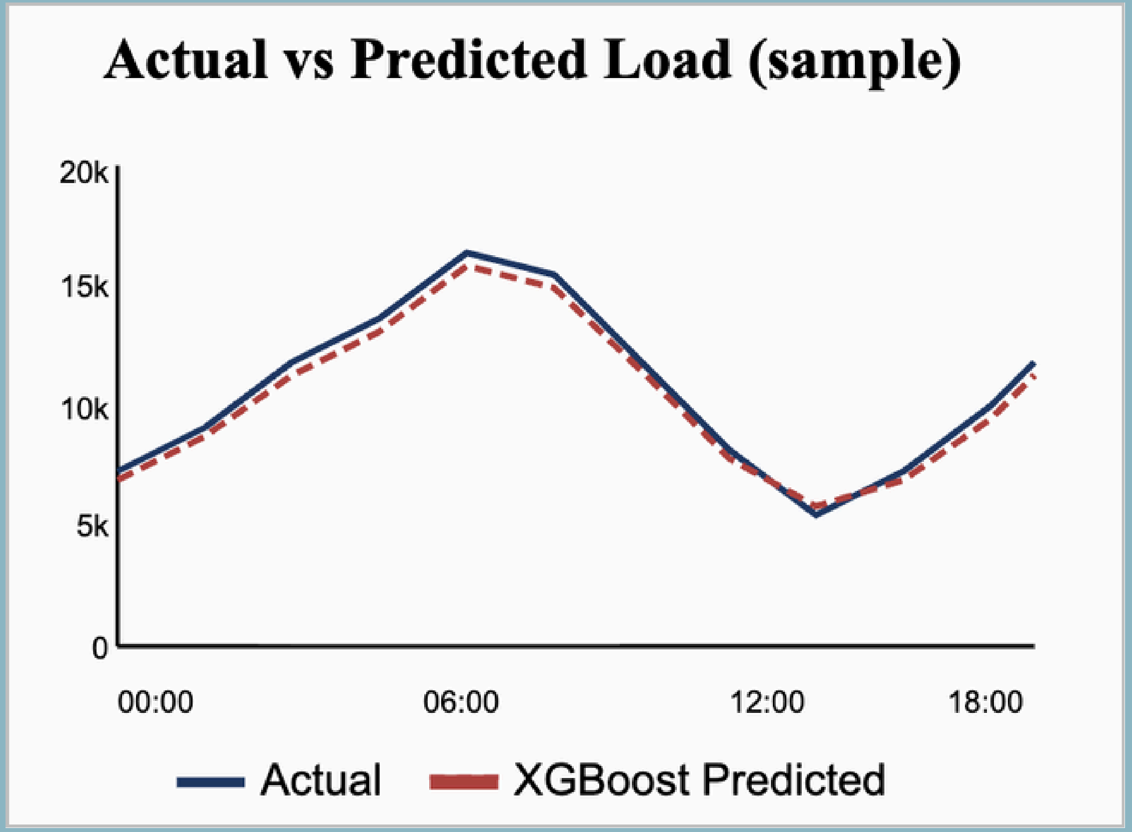
Key Observation

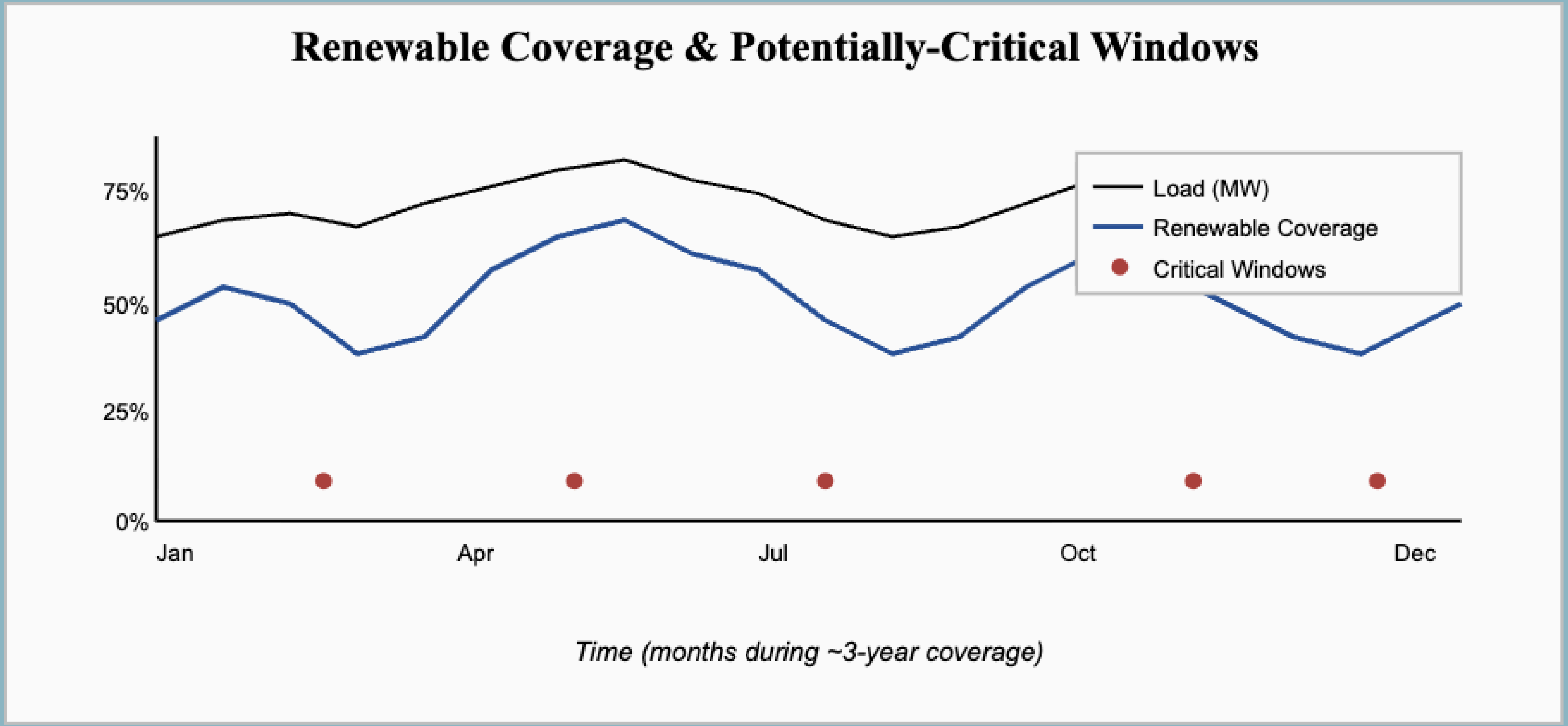
The top three features are lag 1, rolling mean 4 and lag 2 of load. Capture of recent demand history drives forecast accuracy.

RESULTS VISUALIZATION









XGBoost Feature Importance (Top 3)

load_lag_1 50.66%	load_roll_mean_4 24.84%	load_lag_2 23.31%

194 Potentially-Critical Reliability Windows
Analytical demand–renewable windows · not reported as outages

All visuals derived from actual project outputs (chronological evaluation on 20,909 test observations).

RESEARCH AND REFERENCES

Research Context

Existing studies commonly investigate:

- Electricity-load forecasting
- Peak-demand prediction
- Machine-learning forecasting

Our Extension

Load Forecasting + **Renewable Adequacy**

Demand–Renewable Mismatch + **Reliability-Window ID**

References

[1] M.-R. Kazemzadeh, A. Amjadian, and T. Amraee, "A hybrid data mining driven algorithm for long term electric peak load and energy demand forecasting," Energy, vol. 204, 2020.

[2] M. Kumar and N. Pal, "Machine Learning-based Electric Load Forecasting for Peak Demand Control in Smart Grid," Computers, Materials & Continua, vol. 74, no. 3, 2023.

[3] ENTSO-E, "Transparency Platform."

Forecasting + Renewables + Reliability-Window ID — extending prior load-forecasting work toward renewable-adequacy analysis.

CONCLUSION AND FUTURE SCOPE

Conclusion

✓ Dataset successfully validated

✓ ML forecasting pipeline implemented

✓ Linear Regression, Random Forest & XGBoost evaluated

✓ Persistence baseline established

✓ XGBoost achieved the best forecasting performance

✓ Renewable coverage & generation gap analyzed

✓ 194 potentially critical windows identified

Main Takeaway

Forecast Demand + Measure Renewable Support

↓

Identify Potentially Critical Conditions

Future Scope

- Weather-based forecasting
- LSTM / GRU / Transformer
- Probabilistic forecasting
- Real-time monitoring dashboard
- Multi-region analysis
- Energy-storage optimization

Final Message

Data → Prediction → Renewable Analysis → Reliability Insight

THANK YOU · QUESTIONS & DISCUSSION